NHAN TRUONG

Department of Aerospace Engineering
University of Cincinnati, 745C Baldwin Hall, 2850 Campus Way, Cincinnati, Ohio 45221-0070
truongna@mail.uc.edu | (513)-537-2124 | linkedin.com/in/jason-nhan-truong | jasontruong2707.github.io

EDUCATION

University of Cincinnati, Cincinnati, OH Aug 2022 - May 2027 (expected)

- **Bachelor of Science** in Aerospace Engineering **Cumulative GPA:** 3.97/4.0
- **Relevant courses**: Thermodynamics, Aerodynamics, Compressible Flow, Advanced Compressible Flow, Airbreathing Propulsion, Combustion, Aerospace CFD, Rocket Propulsion, Solid Mechanics I, Aerostructures
- **Project:** "*Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows*"
- **Advisor:** Prashant Khare

RESEARCH INTERESTS

- Shock-Droplet and Shock-Bubble Interactions in High-Speed Flows
- Spray Combustion and Flame Stabilization in High-Speed Propulsion Systems
- Large Eddy Simulation of Turbulent Reacting/non-Reacting Flows
- Atomization of Liquid/non-Liquid Propellant
- Machine Learning-accelerated CFD

RESEARCH AND ACADEMIC EXPERIENCE

Undergraduate Research Assistant Jan 2024 - Present
University of Cincinnati Cincinnati, Ohio

1. Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows

- Advisor: Prashant Khare, Achyut Panchal
- Performed a literature review on shock-droplet and shock-bubble interaction, and identified a gap in how the number, location, and size of embedded gas cavities control droplet fragmentation
- Implemented the Saurel six-equation multiphase model in JAX-Fluids, together with non-reflecting boundary conditions and a hyperbolic tangent function that defines a smooth initial interface between phases
- Built the simulation framework and validated it against three prior experimental and numerical studies of shock-droplet interaction, then configured cases that vary the cavity parameters
- Analyzing interface width and span, baroclinic torque generation, and transmitted wave structure inside the droplet with Python post-processing tools to explain how cavity collapse drives fragmentation

2. Fuel Injection, Atomization, and Flame Characteristics in a Model Ramjet Combustor

- Advisors: Prashant Khare, Kamin Manu
- Post-processed large-scale LES data of liquid fuel injected into the crossflow of ramjet combustor with ParaView to study flame stabilization behind a V-gutter flameholder and the sources of flame unsteadiness
- Performed FFT analysis to velocity and pressure time series to identify vortex shedding frequencies and longitudinal acoustic modes driving combustion instability
- Developed Python image processing scripts with OpenCV to extract flame boundaries from instantaneous mean temperature fields and compute mean flame width and its RMS fluctuations
- Implemented POD on 3D temperature snapshot to extract dominant flame modes and relate their frequencies to vortex shedding and acoustic oscillations in the combustor

3. Gel Propellant Atomization

- Advisor: Prashant Khare
- Built a non-Newtonian fluid simulation pipeline in MFC by implementing a Herschel-Bulkley viscosity model in Fortran, and verified it within 1 percent of the analytical Poiseuille velocity profile and within 2 percent of published vortex locations in shear-thinning lid-driven cavity flow
- Performed injection tests with Newtonian and non-Newtonian fluids at different density ratios, and comparing multiphase 5 and 6 equation models to handle the pressure jump across the liquid-gas interface while keeping the solver stable
- Validating the pipeline against an impinging - jet experiment to characterize mixing efficiency, sheet formation, primary breakup mechanisms, and droplet size distributions under variation of ambient pressure

4. Learned WENO Reconstruction for Compressible Turbulence

- Advisor: Prashant Khare
- Building a solver-in-the-loop ML pipeline in JAX-Fluids that learns WENO3 reconstruction weights in a differentiable Euler solver, with the goal of DNS-level accuracy on coarse grids for compressible turbulence
- Training a JAX neural network on spectrally filtered 512 by 512 DNS data of 2D compressible turbulence at a turbulent Mach number of 0.4, with a loss combining MSE, spectral energy, and weight regularization
- Benchmarking the learned scheme (WENO3-NN) against WENO3-JS, WENO5-Z, and ALDM on a 64 by 64 grid using energy spectra, TKE decay, and vorticity correlation with DNS
- Reduced the TKE error from about 40 percent with WENO3-JS to about 3 percent with WENO3-NN after six turbulence time scales, matching the accuracy of WENO5-Z and ALDM

5. Large Eddy Simulations of Recirculation Structures in Isothermal Swirling Coaxial Jet

- Advisor: Kamin Manu
- Examining LES velocity data of three swirler configurations in ParaView and Python to characterize how the recirculation zone changes from pre-vortex breakdown to a central toroidal recirculation zone
- Calculating swirl number, circulation, and swirl momentum factor to quantify how far angular momentum penetrates the recirculation zone core
- Computing mean velocity profiles and turbulence statistics, including TKE and Reynolds stresses, to evaluate mixing and locate recirculation zone boundaries
- Implementing POD and DMD in Python to extract the dominant coherent structures and their frequencies, and to identify the flow instabilities that accompany the transition

RESEARCH PUBLICATIONS

Manuscripts in preparation

1. **Nhan Truong**, Jacob Gamertsfelder, Achyut Panchal, Prashant Khare. “Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows”. Targeted journal: *Physics of Fluids or Journal Fluid Mechanics*
2. Kamin Manu, **Nhan Truong,** Prashant Khare. “Fuel Injection, Atomization, and Flame Characteristics in a Model Ramjet Combustor”. Targeted journal: *Journal of Propulsion and Power*
3. Jeremy Redding, Tripathi Mitansh, Jacob Gamertsfelder, Karmat Siddharth, **Nhan Truong**, Prashant Khare. “High-Temperature Multiphase Dynamics: Shock-Droplet Interaction in Volatile and Metallic Fluids”. Target *Journal of Fluid Mechanics*
4. **Nhan Truong**, Kamin Manu. “Large Eddy Simulations of Recirculation Structures in Isothermal Swirling Coaxial Jet”. Target journal: *Physics of Fluids*

Conference Proceedings/Presentations

1. **Nhan Truong**, Jacob Gamertsfelder, Prashant Khare. “Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows”. *16th Dayton Engineering Sciences Symposium Conference*, Dayton, U.S.A, October 2024

PROFESSIONAL EXPERIENCE

CFD/ CAE Engineering Intern May 2023 - Nov 2023

MiSmart | UAVs/Drones Design & Manufacturing Start-up Southeast Asia

- Established the company's aerodynamic simulation workflow in ANSYS Fluent, and analyzed more than 20 airfoils and 10 aircraft fuselages to check performance against design targets before fabrication
- Designed quadcopter UAV components in SolidWorks and refined them through iterative ground testing, which reduced production cost by 15 percent and extended flight time by 10 percent
- Provided aerodynamic data to the electrical, AI, design, and testing teams to support the design of a LiDAR object-scanning system

LEADERSHIP & TEAM PROJECTS

Programming Chair May 2025 - Present

American Institute of Aeronautics and Astronautics at University of Cincinnati Cincinnati, Ohio

- Directing the planning of more than 17 events per academic year, including faculty panels, industry sessions, and tours of the Air Force Museum and GE Aerospace facilities
- Maintaining relationships with companies such as Gulfstream, GE Aerospace, Rolls-Royce, and AFRL to bring professionals and faculty to technical workshops and speaking events
- Organized sponsor travel for 15 UC students to attend AIAA SciTech 2026, from registration to lodging

Manufacturing and Testing Engineer Aug 2022 - Apr 2023

AeroCats | Remote Control Aircraft Design & Manufacturing Club Cincinnati, Ohio

- Manufactured four RC aircraft with 16-foot wingspans for SAE Aero Design competitions, and laser-cut and assembled more than 1,000 hardwood, pine, and balsa parts
- Measured rotational speed and thrust of six propeller models with three test methods, and compared the results to select the propulsion configuration
- Calculated maximum payload capacity from the thrust data before flight testing, and the team placed fourth overall in the Regular Class at SAE Aero Design West

HONORS & AWARDS

- Davis Professor R.T. Award for the Best Undergraduate Student in Computational Research Mar 2026
- Best Presentation Award in Combustion at 16th Dayton Engineering Sciences Symposium May 2025
- Undergraduate Research Fellowship, University of Cincinnati Oct 2024
- UC Innovation Challenge Scholarship Jan 2023 - Dec 2024
- Fourth Overall in the Regular Class Competition at SAE Aero Design West April 2023
- University Merit Scholarship, University of Cincinnati Aug 2022 - May 2027

TECHNICAL SKILLS

- **Programming:** Fortran, Python, C++, Bash, C, MATLAB, LabVIEW, Visual Basic for Application
- **High-Performance Computing:** SLURM, Linux
- **CFD:** MFC, JAX-Fluids, ECOGEN, Gerris, FluidX3D, STAR-CCM+, ANSYS Fluent, Converge
- **Machine Learning:** JAX, PyTorch, scikit-learn
- **Data Visualization/ Acquisition:** ParaView, Tecplot 360, Blender, POV Ray
- **CAD Modeling:** SolidWorks, Fusion 360

REFERENCES

- Dr. Prashant Khare, Ph.D. (Undergraduate supervisor)

Bradley Jones Chair & Professor, Department of Aerospace Engineering, University of Cincinnati, Cincinnati, Ohio, U.S.A.

Associate Fellow American Institute of Aeronautics and Astronautics (AIAA)

Email: Prashant.Khare@uc.edu; Phone: (513)-556-5248

- Dr. Achyut Panchal, Ph.D. (Undergraduate supervisor)

Assistant Professor, Department of Aerospace Engineering, University of Cincinnati, Cincinnati, Ohio, U.S.A.
Email: achyut.panchal@uc.edu; Phone: (513)-556-3366

- Dr. Kamin Manu, Ph.D. (Undergraduate advisor)

Associate Researcher, Department of Aerospace Engineering, University of Cincinnati, Cincinnati, Ohio, U.S.A.
Email: kaminms@ucmail.uc.edu; Phone: (513)-556-3548